

WIRSCHING'S CONJECTURE 3, AND THE PERIODIC CORRECTION TO BERG-KRÜPPEL'S DENSITY ASYMPTOTIC

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ABSTRACT. Wirsching's 2003 argument for positive predecessor density under the $3n + 1$ map reduces to three conjectures. Conjecture 3 concerns the near-zero asymptotics of an invariant density φ , compared against an explicit asymptotic φ_0 due to Berg and Krüppel. We show that the exact log-Laplace transform of φ decomposes as a smooth part plus a periodic correction H , with closed-form Fourier coefficients in Γ and ζ . This correction is exactly Berg and Krüppel's own undetermined periodic factor from their 1998 saddlepoint asymptotic. On Wirsching's own comparison class, H is constant, and Conjecture 3 holds, with an explicit limit. Off that class, H genuinely oscillates: the stronger, unrestricted asymptotic $\varphi(t) \sim c\varphi_0(t)$ as $t \rightarrow 0^+$ is false. We verify directly that Wirsching's actual requirement, weaker than the literal statement of Conjecture 3, holds with a wide margin.

1. INTRODUCTION

Let T denote the $3n + 1$ map on the positive integers, $T(n) = n/2$ for n even and $T(n) = (3n + 1)/2$ for n odd. Wirsching [1] studies the density of predecessor sets under T through an averaging construction on the 3-adic integers: a family of operators S_l , built from path counts in the predecessor graph, converges in the strong operator topology to a limiting operator S_∞ whose essential part is a Markov chain on $[0, 1]$. That chain has a unique invariant density φ , and Wirsching's proof of positive predecessor density reduces to three conjectures about φ and related quantities, stated as three unbridged gaps in the argument.

The load-bearing one is Conjecture 3. It concerns the behaviour of φ as its argument tends to 0, compared against an explicit comparison function φ_0 . Berg and Krüppel [2] had already produced φ_0 five years earlier, by Laplace transform and saddlepoint analysis, as the asymptotic solution of a truncated version of the functional equation φ satisfies. They did not prove that φ and φ_0 are asymptotically comparable; they only argued for it informally, and left the exact multiplicative constant, and even whether such a constant exists as a genuine limit, open. Wirsching considers the strongest possible reading of that gap directly:

“That replacement would be possible, if, e.g., $\varphi \sim c\varphi_0$ for some constant $c > 0$. It turns out that the following (slightly weaker) conjecture ... suffices.”

He then states Conjecture 3: along a specific comparison class of sequences (z_l) tending to 0 at the borderline rate $z_l \sim l \cdot 3^{-l}$, the ratio $\varphi(z_l)/\varphi_0(z_l)$ converges to a positive constant, uniformly on the class. This is weaker than the unrestricted asymptotic he first considers, precisely because the constant c cancels out of the one inequality his own proof actually needs.

Neither statement had been settled. This paper settles both, in opposite directions.

Theorem 1 (Conjecture 3). *Wirsching's Conjecture 3 is true. On his comparison class $\tilde{A}_\delta$, the ratio $\varphi(z_l)/\varphi_0(z_l)$ converges uniformly to $e^{H(0)} = 0.534122\dots$, where H is the periodic function of Theorem 4 below and the normalization of φ_0 is Berg and Krüppel's own, equation (9.6) of [2].*

Theorem 2 (Failure of the unrestricted asymptotic). *The stronger asymptotic $\varphi(t) \sim c\varphi_0(t)$ as $t \rightarrow 0^+$, for any single constant c and with no restriction to a comparison class, is false. The ratio $\varphi(t)/\varphi_0(t)$ oscillates indefinitely as $t \rightarrow 0^+$, with period $\log 3$ in $\log(1/t)$ and amplitude bounded below by a positive, closed-form constant.*

The two theorems have a single common source. Write $t = e^{-\tau}$ and let $H(w)$ be the $\log 3$ -periodic function appearing in the exact decomposition of $\log \varphi$'s log-Laplace transform (Theorem 4). Wirsching's comparison class fixes τ 's phase modulo $\log 3$ at a single value; every other class of comparison sequences sweeps through all phases. A convergent limit or a genuine oscillation is therefore not a property of φ alone: it depends on which sequences one is allowed to compare along, and the two theorems above are the two readings of the same object at two different scopes.

Corollary 3. *H is exactly Berg and Krüppel's own undetermined periodic factor Q from Proposition 9.3 of [2], for the eigenfunction φ , given here in closed form for the first time.*

The remainder of the paper is organized as follows. Section 2 fixes notation and recalls the exact functional equation φ satisfies. Section 3 constructs H and proves its basic analytic properties, including a rigorous certificate that it is not constant. Section 4 proves a uniform saddlepoint approximation to φ ("formula (A)"), the technical heart of the paper. Section 5 bridges that approximation, built from a smooth auxiliary system, back to φ itself. Section 6 identifies the resulting smooth part exactly with Berg and Krüppel's own saddlepoint expression, proving Corollary 3 and pinning down every constant in closed form. Section 7 assembles the pieces into proofs of Theorems 1 and 2, and checks directly that Wirsching's actual requirement holds with a wide margin. Section 8 states the scope of these results precisely: they resolve Conjecture 3 alone, one of Wirsching's three conjectures. Conjectures 1 and 2 are untouched, and the full positive-predecessor-density theorem is not established here.

2. THE INVARIANT DENSITY AND ITS LAPLACE TRANSFORM

Let $U_1, U_2, \dots$ be independent, identically distributed uniform random variables on $[0, 1]$, and set

$$X = \sum_{j \geq 1} \frac{2}{3^j} U_j \in [0, 1].$$

The self-similar identity $X = \frac{2}{3}U_1 + \frac{1}{3}X'$, with X' an independent copy of X , shows that the density φ of X satisfies

$$\varphi(x) = \frac{3}{2} \int_{3x-2}^{3x} \varphi,$$

equivalently $\varphi'(x) = \frac{9}{2}\varphi(3x)$ on $(0, \frac{2}{3})$: this is Wirsching's invariant density (up to the reflection $x \mapsto 1 - x$ fixing the convention), a translate of Rvachev's atomic function h_3 , and the $a = 3$, $\lambda = 2/3$ case of Berg and Krüppel's family $\lambda g'(t) = a(g(at) - g(at - a + 1))$. It is C^∞ , supported on $[0, 1]$, and constant, equal to $3/2$, on the middle third $[1/3, 2/3]$.

For $s > 0$ write

$$K(s) = \log \mathbb{E}[e^{-sX}] = \sum_{j \geq 1} g\left(\frac{2s}{3^j}\right), \quad g(b) = \log \frac{1 - e^{-b}}{b}.$$

The branch of g is the one analytic on $\{\operatorname{Re} w > 0\}$ and real on $(0, \infty)$; since $(1 - e^{-w})/w$ is analytic and non-vanishing there (its zeros lie on $2\pi i\mathbb{Z} \setminus \{0\}$, and $w = 0$ is a removable singularity with value 1), and the half-plane is simply connected, g exists and is unique. Differentiating termwise, justified because $g(w) = -w/2 + O(w^2)$ as $w \rightarrow 0$ so the tail of the series is normally convergent together with every derivative,

$$t(s) := -K'(s), \quad V(s) := s^2 K''(s) > 0,$$

the second an exact positivity, $K''(s)$ being the variance of X under its exponential tilt by e^{-sX} . As s ranges over $(0, \infty)$, $t(s)$ decreases strictly from $1/2$ to 0 (proved in Lemma 13 below), so studying $\varphi(t(s))$ as $s \rightarrow \infty$ is the same as studying $\varphi(t)$ as $t \rightarrow 0^+$.

Berg and Krüppel's comparison function solves the truncated equation $\lambda g'(t) = a g(at)$ by Laplace transform and saddlepoint analysis. In their normalization (their equation (9.6), with $\alpha = \frac{1}{2} - \frac{\log 2}{\log 3}$, $\beta = \frac{1}{2\log 3}$, and $\gamma, \delta, \varepsilon$ determined by α and β through their Proposition 9.1),

$$\varphi_0(t) = \frac{(2\beta)^\varepsilon}{\sqrt{2\pi}} t^\gamma (-\log t)^\delta \exp\left(-\beta \log^2 \frac{t}{-\log t}\right).$$

We write $\varphi_{0,\text{bare}}$ for the same expression without the leading constant $(2\beta)^\varepsilon/\sqrt{2\pi}$; this is the normalization used in this project's earlier numerical work, and the two differ only by that constant.

3. THE PERIODIC CORRECTION

Set $c = \log 3$ and write $w = \log s$, $L(w) = K(e^w)$. Substituting into the defining series and splitting off the $j = 1$ term against the rest reproduces the recurrence

$$(1) \quad L(w+c) - L(w) = \log(1 - e^{-2e^w}) - w - \log 2.$$

Let

$$Q(w) = -\frac{w^2}{2c} + \left(\frac{1}{2} - \frac{\log 2}{c}\right)w,$$

chosen so that $Q(w+c) - Q(w) = -(w+a) + c/2$ with $a := \frac{1}{2} - \frac{\log 2}{c}$, matching the leading terms of (1) exactly.

Theorem 4. *Define*

$$(2) \quad H(w) := L(w) - Q(w) + \sum_{k \geq 0} \log(1 - e^{-2 \cdot 3^k e^w}).$$

Then H is exactly c -periodic, $H(w+c) = H(w)$ for every $w \in \mathbb{R}$, and

$$L(w) = Q(w) + H(w) + E(w), \quad E(w) = -\sum_{k \geq 0} \log(1 - e^{-2 \cdot 3^k e^w}),$$

with $E^{(j)}(w) = O_j(e^{jw-2e^w})$ for every $j \geq 0$: doubly exponentially small, together with all its derivatives, as $w \rightarrow +\infty$.

Proof. Write $d(w) = \log(1 - e^{-2e^w})$, so (1) reads $L(w+c) - L(w) = d(w) - w - \log 2$, and $Q(w+c) - Q(w) = -(w+a) + c/2$. Since $-\log 2 = -a - c/2$ by the definition of a ,

$$(L - Q)(w+c) - (L - Q)(w) = d(w) - w - \log 2 + w + a - c/2 = d(w).$$

The series in (2) telescopes this exactly: with $D(w) := (L - Q)(w)$,

$$D(w) + \sum_{k \geq 0} d(w+kc) = D(w+c) - d(w) + \sum_{k \geq 0} d(w+kc) = D(w+c) + \sum_{k \geq 0} d(w+(k+1)c),$$

so the right side of (2) is invariant under $w \mapsto w+c$, that is, H is c -periodic. Convergence of the series and its termwise differentiability, together with the stated bound on $E^{(j)}$, follow from $d(w) = O(e^{-2e^w})$ and the corresponding bounds on its derivatives, since $2 \cdot 3^k e^w \geq 2e^w$ for $k \geq 0$ makes the series dominated, uniformly for w in any half-line bounded away from $-\infty$, by a geometric series in 3^{-k} inside the exponent. $\square$

The decomposition $L = Q + H + O(1)$ (in the limit sense; exactly $L = Q + H + E$ with E negligible) is unique: any two periodic functions differing from $L - Q$ by an $o(1)$ term as $w \rightarrow \infty$ must be equal, being periodic functions that agree in the limit along every residue class of w modulo c .

Proposition 5. *H has the Fourier expansion $H(w) = \sum_{m \in \mathbb{Z}} \hat{H}(m) e^{i\omega_m w}$, $\omega_m = 2\pi m/c$, with*

$$\hat{H}(0) = \frac{\log 2}{2} - \frac{\log^2 2}{2c} - \frac{c}{12} - \frac{A}{c}, \quad A = \frac{\pi^2}{12} - \frac{\gamma_E^2}{2} - \gamma_1,$$

$$\hat{H}(m) = -\frac{2^{i\omega_m}}{c} \Gamma(-i\omega_m) \zeta(1 - i\omega_m), \quad m \neq 0,$$

where γ_E is the Euler–Mascheroni constant and γ_1 the first Stieltjes constant.

Sketch. The Mellin transform of g on the strip $-1 < \operatorname{Re} z < 0$ is $g^*(z) = -2^{-z} \Gamma(z) \zeta(1+z)$, obtained by integrating by parts onto g' and using the classical continuation $\int_0^\infty (\frac{1}{e^u-1} - \frac{1}{u}) u^{z-1} du = \Gamma(z) \zeta(z)$. Substituting the defining series for K and summing the resulting geometric series in the Mellin variable gives the Mellin transform of L ; shifting the inversion contour past the triple pole at the origin and past each simple pole at $z = i\omega_m$ reproduces $Q + \hat{H}(0)$ and the Fourier modes above, respectively. $\square$

The series for H' and H'' converge absolutely and uniformly, since $|\Gamma(-i\omega_m)| = \sqrt{\pi/(\omega_m \sinh \pi \omega_m)}$ decays exponentially in m while $\zeta(1-i\omega_m)$ grows only polynomially, and this yields explicit, certified bounds:

$$(3) \quad \sup_w |H'(w)| \leq 0.0011977472315550332, \quad \sup_w |H''(w)| \leq 0.0068518962896650951.$$

Proposition 6 (Certified non-constancy). *H is not constant. Rigorously,*

$$-0.000377190280943987 < H(0) - H(\log(3/2)) < -0.000377190280943985,$$

and consequently $\operatorname{osc}(H) := \sup H - \inf H \geq 3.7719 \times 10^{-4}$; a full optimization gives the certified enclosure $\operatorname{osc}(H) \in [4.18744947692 \times 10^{-4}, 4.18756644224 \times 10^{-4}]$.

Sketch. Write $g(b) = -b + \log S(b)$, $S(b) = \sinh(b)/b = \sum_{n \geq 0} b^{2n}/(2n+1)!$. Termwise domination of $6^n n! \leq (2n+1)!$ gives $0 \leq \log S(b) \leq b^2/6$ for $b \geq 0$, hence an explicit, two-sided enclosure of the tail $\sum_{r > R} g(s/3^r)$ for any R and $s > 0$; combined with a finite, rigorously evaluated head (interval arithmetic, ball arithmetic in Γ and ζ , or direct high-precision evaluation of (2)), this certifies the stated interval to arbitrary precision. The full oscillation is certified the same way, via the Lipschitz bound in (3) and a fine grid search over one period. $\square$

4. A UNIFORM SADDLEPOINT APPROXIMATION

Define, for $s > 0$,

$$\varphi_{\text{sp}}(t(s)) := \frac{\exp(K(s) + st(s))}{\sqrt{2\pi K''(s)}}.$$

This section proves that $\varphi(t(s))/\varphi_{\text{sp}}(t(s)) \rightarrow 1$ as $s \rightarrow \infty$, uniformly over every phase of s modulo $3\mathbb{Z}$, which is what makes the two theorems of Section 7 depend on different scopes rather than on different mathematics.

For $n \geq 2$, define the derivative-normalized functions $\kappa_n(w) = (-1)^n w^n g^{(n)}(w)$ on $\{\operatorname{Re} w > 0\}$; $\kappa_n(b)$, $b > 0$, is b^n times the n -th cumulant of the random variable with density $b e^{-bw}/(1 - e^{-b})$ on $[0, 1]$. Writing $x = w/2$,

$$\kappa_2(w) = 1 - \frac{x^2}{\sinh^2 x}, \quad \kappa_3(w) = 2 - \frac{2x^3 \cosh x}{\sinh^3 x}, \quad \kappa_4(w) = 6 + \frac{2x^4(1 - 3 \coth^2 x)}{\sinh^2 x},$$

and from the Bernoulli expansion $g(w) + w/2 = \sum_{k \geq 1} B_{2k} w^{2k}/(2k(2k)!)$ on $|w| < 2\pi$,

$$\kappa_n(w) = (-1)^n \sum_{k \geq \lceil n/2 \rceil} \frac{B_{2k} w^{2k}}{2k(2k-n)!}, \quad n \geq 2,$$

so that $\kappa_n(w) = O(w^2)$ as $w \rightarrow 0$ for every $n \geq 2$.

Write $s = \rho \cdot 3^N/2$ for the unique $N = \lfloor \log_3(2s) \rfloor \in \mathbb{Z}$ and $\rho \in [1, 3)$, and $b_j = 2s/3^j$. Then $\{b_j : j \geq 1\} = \{\rho \cdot 3^m : m \leq N-1\}$: exactly N of these, the smallest being $\rho \geq 1$, are ≥ 1 ; the rest form a geometric tail below 1. This single structural fact is the source of every phase-independent constant below.

Lemma 7 (Boundedness off the real axis). *For $w = b(1+iu)$, $b > 0$, $|u| \leq 1$, with $e(r) = r^2/\sinh^2 r$ and $f(r) = r^3 \cosh r/\sinh^3 r$,*

$$|\kappa_2(w)| \leq 1 + 2e(b/2), \quad |\kappa_3(w)| \leq 2 + 2 \cdot 2^{3/2} f(b/2).$$

Both bounds are finite over the whole sector $|\operatorname{Im} w| \leq \operatorname{Re} w$; $M_2 := \sup |\kappa_2| \leq 3$, $M_3 := \sup |\kappa_3| \leq 2 + 2^{5/2} = 7.657$. In addition, for $|w| \leq 2$, $|\kappa_2(w)| \leq 0.114|w|^2$ and $|\kappa_3(w)| \leq 0.0119|w|^4$.

Proof. $|\sinh z|^2 = \sinh^2(\operatorname{Re} z) + \sin^2(\operatorname{Im} z) \geq \sinh^2(\operatorname{Re} z)$ and $|\cosh z|^2 \leq \cosh^2(\operatorname{Re} z)$; the first excludes the poles of $\coth$, since $\operatorname{Re}(w/2) = b/2 > 0$ never approaches $\pi i k$. This gives the sector bounds directly. Monotonicity of e is immediate from the increasing power series of $\sinh(r)/r$; monotonicity of f is Lazarević's inequality, $(\sinh r/r)^3 > \cosh r$, checked by comparing alternating truncated series near $r = 0$ and by monotonicity of $\coth r - 1/r$ away from it. The local bounds at $|w| \leq 2$ follow from a Cauchy estimate on $|w| = 2$ using $|B_{2k}| = 2(2k)! \zeta(2k)/(2\pi)^{2k}$ and $\zeta(2k) \leq \zeta(2)$, respectively $\zeta(4)$. $\square$

Lemma 8 (Variance). *For all $\rho \in [1, 3)$, $N \geq 1$, with $V := s^2 K''(s) = \sum_j \kappa_2(b_j)$,*

$$N - A \leq V \leq N + 0.1283, \quad A := \sum_{m \geq 0} e(3^m/2) = 1.4269413069 \dots$$

Proof. $\kappa_2(b) = 1 - e(b/2) \in (0, 1)$. The lower bound drops the N terms with $m \geq 0$ (each < 1 , contributing at most N) below their true value and adds back the $m < 0$ tail, maximal at $\rho = 1$ since e decreases; the upper bound bounds the same tail via Lemma 7's local estimate. $\square$

Lemma 9 (Tail bound and existence of the density). *If $b_0 \geq 1$ and $b_k = 3^k b_0$, then $\sum_{k \geq 0} \log \coth(b_k/2) \leq 3e^{-b_0}$. Consequently, with $\Lambda(y) := K(s(1+iy)) - K(s) + isyt(s)$,*

$$|e^{\Lambda(y)}| \leq e^{3/e} (1+y^2)^{-N/2} \leq 3.0152 (1+y^2)^{-N/2}$$

for every real y . In particular X has a continuous density, the tilted characteristic function is integrable for $N \geq 3$, and Fourier inversion at $x = t(s)$ gives

$$(4) \quad R(s) := \frac{\varphi(t(s))}{\varphi_{\text{sp}}(t(s))} = \sqrt{\frac{V}{2\pi}} \int_{\mathbb{R}} e^{\Lambda(y)} dy,$$

a real quantity, since $\Lambda(-y) = \overline{\Lambda(y)}$.

Proof. $\log \coth(x/2) = 2 \operatorname{artanh}(e^{-x}) \leq 2e^{-x}/(1 - e^{-2x})$; summing over the geometric orbit $b_k = 3^k b_0$ and bounding the resulting series by its first few terms gives the stated constant. X 's density is continuous because the sum of the first two summands already has a continuous, compactly supported density and convolution preserves continuity. For Λ : $t(s) = -K'(s)$ cancels the linear term of the tilted transform exactly, and each remaining factor $e^{g(b_j(1+iy)) - g(b_j)}$ has modulus at most 1 (a tilted characteristic function), with modulus at most $\coth(b_j/2)/\sqrt{1+y^2}$ on the N indices where $b_j \geq 1$; the tail estimate applies to exactly that geometric set, with $b_0 = \rho \geq 1$. $\square$

Lemma 10 (Central expansion). *With $h(u) = g(b(1+iu))$, Taylor's theorem in integral form (the Lagrange form is invalid for a complex-valued function of a real variable) gives*

$$h(y) = h(0) + h'(0)y + h''(0)\frac{y^2}{2} + \frac{1}{2} \int_0^y (y-u)^2 h'''(u) du.$$

Consequently, for $a \in (0, 1]$,

$$S := \sum_j \sup_{|u| \leq a} |h_j'''(u)| \leq 2N + 3.7442,$$

and hence $|\Lambda(y) + Vy^2/2| \leq B|y|^3$ for $|y| \leq a$, with $B := S/6 \leq (2N + 3.7442)/6$.

Proof. The bound on h_j''' follows from Lemma 7 applied to κ_3 , summed over the N indices with $b_j \geq 1$ (via f 's monotonicity, worst case $\rho = 1$) and the remaining geometric tail (via the local bound). $\square$

Theorem 11 (Formula (A)). *There is an absolute, phase-independent N_0 and an explicit, strictly decreasing function $E(N) = O(N^{-1/2})$ such that $|R(s) - 1| \leq E(N)$ for every s with $N \geq N_0$, uniformly over $\rho \in [1, 3)$. Consequently $\varphi(t) = \varphi_{\text{sp}}(t)(1 + o(1))$ as $t \rightarrow 0^+$, with no restriction on the sequence of t 's.*

Proof. Split (4) at $|y| = a_N = N^{-1/3}$. On the central region, Lemma 10 bounds the cubic remainder by a constant depending only on N , giving a Gaussian integral plus an explicit error of order $N^{-1/2}$ after using $\int_{\mathbb{R}} e^{-Vy^2/2}|y|^3 dy = 4/V^2$ and Lemma 8's bound $V = N + O(1)$; the exponent $1/3$ is chosen so this error term is $o(1)$. On the tail region, Lemma 9's bound integrates to a term of order $N^{-1/6}e^{-N^{1/3}/2}$, which requires $a_N \gg N^{-1/2}$, again satisfied by $\frac{1}{3} < \frac{1}{2}$. Every constant entering E is a supremum over $b > 0$ or a sum over one complete geometric orbit bounded using $\rho \geq 1$, hence independent of ρ ; a concrete choice gives $N_0 = 17$, $E(17) < 1$, $E(N) \leq 4.1 N^{-1/2}$ for all $N \geq 17$, with $\sqrt{N} E(N) \rightarrow \frac{4}{3}e^{1/3}(2\pi)^{-1/2} = 0.742358\dots$. Since $t(s)$ ranges bijectively over $(0, 1/2)$, this is exactly the statement $\varphi(t) = \varphi_{\text{sp}}(t)(1 + o(1))$ as $t \rightarrow 0^+$. $\square$

Remark 12. The rate is not sharp. Carrying the central expansion to fourth order and using the exact parities $V_3 = -2N + O(1)$, $V_4 = 6N + O(1)$ (the third and fourth cumulant sums, by the same argument as Lemma 8) gives the Edgeworth correction $R(s) = 1 - \frac{1}{12N} + O(N^{-3/2})$, matching the classical Stirling correction for a Gamma($N, 1$) law: after tilting, the $b_j \geq 1$ summands behave asymptotically like independent Exp(1) variables, so X under its tilt is asymptotically Gamma($N, 1$)-distributed, and ρ enters only through $O(1)$ shifts of the cumulants, at order N^{-2} . We do not need this sharper rate and do not prove the remainder term here; Theorem 11's qualitative $o(1)$ is all that Section 7 uses.

5. THE ENVELOPE LEMMA

Theorem 11 compares φ to φ_{sp} , a quantity built from the full transform $K = Q + H + E$. It remains to compare φ_{sp} to a smooth auxiliary system built from Q alone, so that the periodic part H can be extracted explicitly. Write $t = e^{-\tau}$, $g_\tau(w) = L(w) + e^{w-\tau}$, $g_{0,\tau}(w) = Q(w) + e^{w-\tau}$, $B(w) = w/c - a$.

Lemma 13 (True and smooth saddles). *For each τ , g_τ has a unique global minimizer $w^*(\tau)$, at which $g_\tau''(w^*) = V(r^*) > 0$, $r^* = e^{w^*}$; and $g_{0,\tau}$ has a unique local minimizer $w_0(\tau)$ among solutions with $\beta := B(w_0) = e^{w_0-\tau} > 1/c$. Both are characterized by $\tau = \Phi(w^*) = \Phi_0(w_0)$ for $\Phi(w) = w - \log B(w)$ (a strictly increasing bijection $\mathbb{R} \rightarrow (\log 2, \infty)$, since $t(s)$ is strictly decreasing) and $\Phi_0(w) = w - \log B_0(w)$, and satisfy*

$$|w^* - w_0| = O(1/\beta), \quad 0 \leq g_\tau(w_0) - g_\tau(w^*) = O(1/\beta).$$

Sketch. Strict convexity of K gives the global uniqueness of w^* . At w_0 , $g_\tau'(w_0) = H'(w_0) + O(e^{-2e^{w_0}})$ by Theorem 4, bounded by (3); since g_τ'' is bounded above and below by explicit constants on $[w_0 - 1, w_0 + 1]$ (again via (3) and Lemma 7), a sign-change argument for g_τ' locates w^* within $O(1/\beta)$ of w_0 , and Taylor's theorem about w^* then bounds the resulting gap in g_τ -values by the same order. $\square$

Proposition 14 (Envelope estimate). *With $S(\tau) := \log \varphi_{\text{sp}}(t) = g_\tau(w^*) + w^* - \frac{1}{2} \log(2\pi V(r^*))$ and $P(\tau) := Q(w_0) + \beta + w_0 - \frac{1}{2} \log(2\pi(\beta - \frac{1}{c}))$,*

$$|S(\tau) - P(\tau) - H(w_0(\tau))| = O(1/\tau) \quad (\tau \rightarrow \infty),$$

uniformly in the phase of τ modulo c .

Sketch. Expanding $S(\tau) - P(\tau) - H(w_0)$ exactly using $L = Q + H + E$ and Lemma 13 produces four terms: the true envelope gap $g_{0,\tau}(w^*) - g_{0,\tau}(w_0)$, which is $O(\beta(w^* - w_0)^2) = O(1/\beta)$; the periodic-part difference $H(w^*) - H(w_0) = O(\eta|w^* - w_0|) = O(1/\beta)$ using (3); the saddle-location shift $w^* - w_0$ itself, $O(1/\beta)$; and a normalization term comparing $V(r^*)$ to $\beta - 1/c$, again $O(1/\beta)$ using (3)'s bound on H'' . Since $\tau = w_0 - \log \beta$, $\beta/\tau \rightarrow 1/c$, and all four terms are $O(1/\tau)$ with constants free of the phase of τ . $\square$

6. THE BERG–KRÜPPEL IDENTITY

Proposition 15. *$P(\tau)$ is identically Berg and Krüppel's own saddlepoint expression for $\log \varphi_0$: with $Y = cB_0$, $r = -ca - \log c$, the saddle equation for φ_0 's defining Laplace inversion is exactly $Y - \log Y = \tau + r$, and $P(\tau) = Q(w_0) + \beta + w_0 - \frac{1}{2} \log(2\pi(\beta - 1/c))$ coincides term for term with Berg and Krüppel's construction in their Proposition 9.1: $2\beta y - \alpha = B_0$, $2\beta y - 2\beta - \alpha = \beta - 1/c$, $\alpha y - \beta y^2 = Q(w_0)$ (in their notation, $y = w_0$).*

Sketch. Direct substitution, using $\alpha = a$, $\beta = 1/(2c)$ from their equation (9.4) applied to $a = 3$, $\lambda = 2/3$. $\square$

Series-reverting $Y - \log Y = \tau + r$ to second order, $Y = \tau + \log \tau + r + (\log \tau + r)/\tau + O(\tau^{-2} \log^2 \tau)$, and substituting into P gives

$$P(\tau) = -\frac{(\tau + \log \tau)^2}{2c} + \left(1 + \frac{1}{c} - \frac{r}{c}\right)\tau + \left(\frac{1}{2} - \frac{r}{c}\right)\log \tau + C_P + O\left(\frac{\log^2 \tau}{\tau}\right),$$

$$C_P = -\frac{r^2}{2c} + r + ca + \frac{ca^2}{2} - \frac{\log 2\pi}{2} + \frac{\log c}{2} = -0.9576743183133982760669122497855855\dots$$

Matching the τ and $\log \tau$ coefficients against Berg and Krüppel's γ , δ (their equations following (9.6)) reproduces both exactly, and

$$C_P = \varepsilon \log(2\beta) - \frac{1}{2} \log(2\pi) = \log \frac{(2\beta)^\varepsilon}{\sqrt{2\pi}},$$

so $P(\tau) - \log \varphi_{0,\text{bare}}(e^{-\tau}) \rightarrow C_P$ and $P(\tau) - \log \varphi_0(e^{-\tau}) \rightarrow 0$: P is the exact log of Berg and Krüppel's normalized, prefactor-included saddlepoint expression, up to an $O(\tau^{-1} \log^2 \tau)$ remainder.

Proof of Corollary 3. By Proposition 14 and 15, $S(\tau) = \log \varphi_0(e^{-\tau}) + H(w_0(\tau)) + O(\tau^{-1} \log^2 \tau)$: the periodic factor multiplying Berg and Krüppel's asymptotic solution, in the exact (not merely truncated) equation for φ , is H . This is exactly the arbitrary c -periodic factor Q their Proposition 9.3 leaves undetermined for the general solution of the untruncated equation, restricted here to the specific eigenfunction φ and given, via Proposition 5, in closed form. $\square$

7. PROOFS OF THE MAIN THEOREMS

Proof of Theorem 1. Wirsching's class $\tilde{A}_\delta$ (his equations (1.5) and (3.2)) consists of sequences (x_l) with $\lfloor 3^l x_l \rfloor = k_l$ satisfying $|l - k_l| \leq \delta \sqrt{l}$ for every l ; hence $\lambda_l := 3^l x_l / l \rightarrow 1$ uniformly at rate $O(\delta/\sqrt{l})$ on the class. Writing $z_l = \lambda_l l 3^{-l}$, $\tau_l = -\log z_l = lc - \log l - \log \lambda_l$, the saddle location satisfies $w_0(\tau_l) = lc - \log \lambda_l + O(1/l)$ by Lemma 13's defining equation, so $w_0(\tau_l) \bmod c \rightarrow 0$ uniformly on the class, at rate $O(1/\sqrt{l})$. By Theorem 11, Proposition 14, and continuity of H (indeed Lipschitz, by (3)),

$$\log \varphi(z_l) - \log \varphi_0(z_l) = H(w_0(\tau_l)) + o(1) \rightarrow H(0)$$

uniformly on $\tilde{A}_\delta$, i.e. $\varphi(z_l)/\varphi_0(z_l) \rightarrow e^{H(0)}$ uniformly. This is exactly Conjecture 3, with the constant given explicitly. $\square$

Proof of Theorem 2. As $t = e^{-\tau}$ ranges over $(0, 1/2)$ with no restriction, $w_0(\tau) \bmod c$ ranges over a full period (since Φ_0 is a bijection and $\tau \mapsto \tau \bmod c$ composed with Φ_0^{-1} is onto one period as $\tau \rightarrow \infty$), so by the same combination of results used above, $\log \varphi(t) - \log \varphi_0(t) = H(w_0(\tau)) + o(1)$ takes values arbitrarily close to both $\sup H$ and $\inf H$ infinitely often. By Proposition 6, $\text{osc}(H) > 0$ rigorously, so $\varphi(t)/\varphi_0(t)$ does not converge, and $\limsup / \liminf \geq e^{\text{osc}(H)} \geq 1.0004188$. $\square$

Proposition 16 (Wirsching's actual requirement). *Wirsching's own use of Conjecture 3, via $\varphi'(x) = \frac{9}{2}\varphi(3x)$ on $(0, \frac{2}{3})$ and Berg and Krüppel's proven limit $\lim(2/3^{l+1})\varphi'_0(x_l^+)/\varphi_0(x_l^+) = 2/3$ uniformly on the class (their equation (7.13)), reduces to*

$$\limsup_l \Lambda_l < \frac{3}{2}, \quad \Lambda_l := \frac{\varphi(3x_l^+)/\varphi_0(3x_l^+)}{\varphi(x_l^+)/\varphi_0(x_l^+)}.$$

This holds with a wide margin: $\Lambda_l \rightarrow 1$, since the phase shift under $t \mapsto 3t$ is $w_0(\tau - c) - w_0(\tau) + c = O(1/\tau)$, phase-independent, so H cancels in the ratio Λ_l to the same order as Proposition 14's error term.

8. DISCUSSION

These results settle Conjecture 3 alone. Wirsching's positive-predecessor-density argument reduces to three conjectures; Conjectures 1 and 2, concerning respectively a pointwise-versus-average transfer for the Elka functions and a uniformity-near-a-singularity statement for the limiting operator S_∞ , are untouched here, and the full theorem is not established by this paper.

The two theorems together explain, rather than merely report, why Wirsching stated the weaker Conjecture 3 instead of the stronger $\varphi \sim c\varphi_0$ he considers first: the stronger statement is false, genuinely and by a mechanism with a closed form, not through some correctable technical looseness. The class on which the weaker statement holds is not an arbitrary technical device either. It is exactly the one phase, among the continuum swept by H , at which the ratio converges, and that phase happens to sit close to H 's minimum: $H(0) - \min H = 5.25 \times 10^{-6}$ against a total oscillation of 4.19×10^{-4} . We do not know whether this is coincidental.

The rate is not established beyond $o(1)$. The sharper Edgeworth correction of Remark following Theorem 11, and the corresponding sharper form of the envelope and Berg–Krüppel comparisons, would give an explicit polynomial rate throughout; nothing in this paper's two theorems requires it.

9. CODE AND DATA AVAILABILITY

Every numerical claim in this paper, including the interval-arithmetic certificate behind Proposition 6, the constants of Theorem 11, and the identity of Proposition 15, is backed by a runnable script in the accompanying repository, <https://github.com/faculdade/wirsching-conjecture3-proof>, organized by section with a README per script stating what it verifies and how to run it.

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REFERENCES

- [1] G. J. Wirsching, *On positive predecessor density in $3n + 1$ dynamics*, Discrete Contin. Dyn. Syst. **9** (2003), no. 3, 771–787.
- [2] L. Berg and M. Krüppel, *On the solution of an integral-functional equation with a parameter*, Z. Anal. Anwendungen **17** (1998), no. 1, 159–181.
- [3] G. J. Wirsching, *The Dynamical System Generated by the $3n + 1$ Function*, Lecture Notes in Mathematics, vol. 1681, Springer, Berlin, 1998.

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